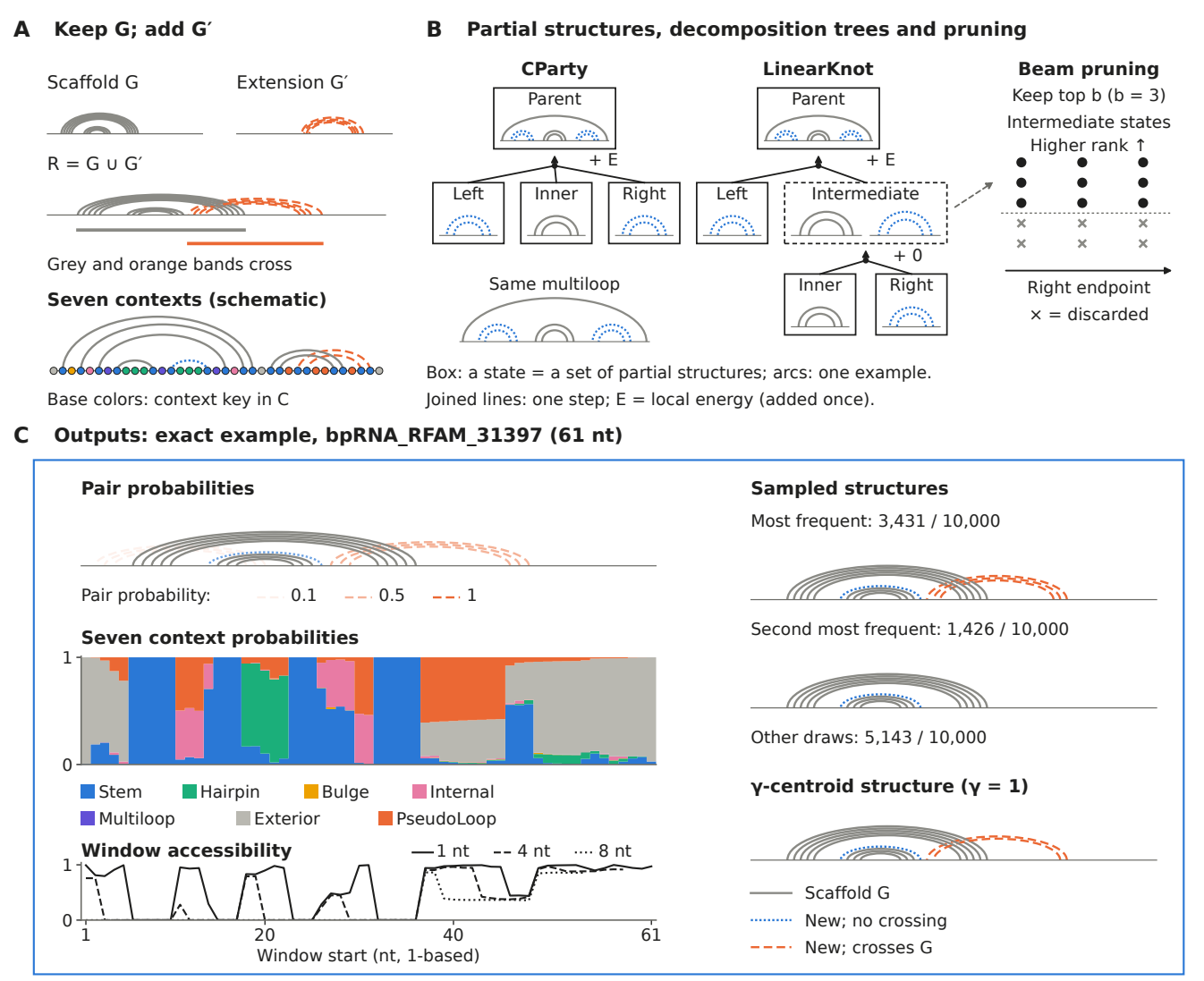
A Keep G; add G′
Scaffold G
Extension G′
R = G ∪ G′
Grey and orange bands cross
Seven contexts (schematic)
Base colors: context key in C
B Partial structures, decomposition trees and pruning
CParty
Parent
+ E
Left
Inner
Right
Same multiloop
LinearKnot
Parent
+ E
Left
Intermediate
+ 0
Inner
Right
Beam pruning
Keep top b (b = 3)
Intermediate states
Higher rank ↑
Right endpoint
× = discarded
Box: a state = a set of partial structures; arcs: one example.
Joined lines: one step; E = local energy (added once).
C Outputs: exact example, bpRNA_RFAM_31397 (61 nt)
Pair probabilities
Pair probability: 0.1 0.5 1
Seven context probabilities
1
0
Stem
Hairpin
Bulge
Internal
Multiloop
Exterior
PseudoLoop
Window accessibility
1 nt
4 nt
8 nt
1
0
1
20
40
61
Window start (nt, 1-based)
Sampled structures
Most frequent: 3,431 / 10,000
Second most frequent: 1,426 / 10,000
Other draws: 5,143 / 10,000
γ-centroid structure (γ = 1)
Scaffold G
New; no crossing
New; crosses G